

Provenance Is Not Validity: The Vacuity of Cryptographically Verified Observation

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Abstract—A growing class of systems anchors scientific measurements on a blockchain to make them trustworthy: observations are signed at the source, time-stamped, and recorded on a distributed ledger, and are said to become provably reliable and reproducible. We show that this claim conflates two guarantees that must be kept apart. Cryptographic anchoring establishes provenance, the authenticity, integrity, and time of a record. It does not, and by construction cannot, establish validity, whether the instrument was calibrated, whether it pointed where it claims, and whether the observation supports a downstream scientific read. A miscalibrated telescope’s output signs and anchors exactly as cleanly as a correct one, so the ledger clears every record with equal confidence regardless of scientific worth. We formalize this through consumer-relative observation and characterize such a system as a false-clear mechanism: it certifies the source, not the reader who must use the data. We then supply the missing layer. A validity attestation re-evaluated at use time, a measured false-clear rate for the network, and re-observation as the consumer test convert provenance into evidence. On real Breakthrough Listen L-band data, an ON-OFF cadence consumer rejects roughly 99.95% of raw single-scan detections as interference, a discrimination no signature provides. By our characterization this 99.95% is the base-rate false-clear that a provenance-only certificate over the same stream inherits. For trusted observational datasets provenance is necessary and not sufficient, and the validity layer, not the ledger, is where trust is earned.

Index Terms—scientific data integrity; provenance versus validity; consumer-relative observation; false-clear rate; technosignature verification; blockchain critique

I. INTRODUCTION

Observational science is acquiring a new trust infrastructure. Systems now sign a measurement at the sensor, time-stamp it, and anchor it on a public ledger, and present the result as data that can be trusted without trusting the operator. Proposals that anchor scientific data provenance on a distributed ledger are numerous and active, contributing signed, time-stamped observations and naming reliable, tamper-proof, and reproducible datasets as the outcome [1].

We make one argument. Cryptographic anchoring proves that a record is authentic, unaltered, and old. It does not prove that the record is right. Those are different guarantees, and only the first is a property of a signature. The second is a property of the instrument, the calibration, and the downstream use, and no amount of anchoring reaches it. A ledger that anchors a miscalibrated telescope’s output has proven the wrong thing well.

We are careful about what the cryptography does earn. For attribution, where the requirement is non-repudiable chain of custody, provenance is exactly right. For pre-registration, where a timestamp that provably predates a claim resists after-the-fact story fitting, provenance is again the right tool. The paper does not dispute these. It disputes the slide from provenance to validity that the reliability and reproducibility framing performs.

The contribution is threefold. First, we separate provenance from validity precisely and locate the conflation in the class of onchain-observation systems (Sections II and III). Second, we characterize such a system, through consumer-relative observation, as a false-clear mechanism, and define a measurable false-clear rate for it (Section IV). Third, we give the validity layer that the provenance layer lacks, and we demonstrate the central part of it, a re-observation consumer, on real Breakthrough Listen data (Sections V and VI).

II. BACKGROUND: ONCHAIN OBSERVATION

The pattern is consistent across the class. A sensor holds a signing key. Each observation, or a digest of it, is signed at the point of capture, associated with a time, and written to or anchored on a distributed ledger, with the bulk data held in decentralized or content-addressed storage [2]. The stated goals are tamper-evidence, non-repudiation, and an auditable history that any party can check without trusting a central operator.

We treat such a system as a pure-provenance mechanism, which is the design these public claims describe. This is the strongest reading, because it takes the mechanism at face value. If a particular specification adds instrument-level checks, the critique in Section III narrows from the

design to the claim, and the constructive layer of Section V still applies.

Cryptographic timestamping and tamper-evident logs are mature and well understood outside the ledger setting [3], [2]. A blockchain is one way to obtain tamper-evident ordering and open verifiability. Whether it is the most economical way is not our question. Our question is what the ordering and the signatures are *about*.

III. TWO GUARANTEES, KEPT APART

Anchoring establishes three facts, and all three concern the bytes. Authenticity: a known key signed this record. Integrity: the record is unchanged since it was signed. Existence at a time: the record existed by a given timestamp. Call the conjunction *provenance*.

Validity is a different property. It asks whether the instrument was calibrated within its known-good envelope, whether it pointed where the metadata claims, whether the exposure was free of saturation, artifacts, and interference, and whether the recorded observation supports the specific read a downstream consumer intends. Call this *validity*.

The two are independent. A saturated sensor, a mis-pointed mount, or an uncalibrated gain produces a record that signs and anchors exactly as cleanly as a correct observation. The signature is a function of the bytes and the key. It is invariant to whether the bytes describe a good measurement. Therefore anchoring cannot separate a valid observation from an invalid one, and a system built only on anchoring clears every record with equal confidence regardless of its scientific worth.

The reliability and reproducibility claims cross this line. Reliability is a validity property of the instrument. Reproducibility is a property of independent re-observation and re-derivation, and a signature never re-observes anything. Only tamper-evidence, the middle term, is a provenance property that the cryptography actually delivers.

IV. A FALSE-CLEAR CHARACTERIZATION

Observation Theory holds that an observation is meaningful only relative to the consumer that reads it, and that a certificate is worth only what it lets that consumer verify [4]. Let a record x carry a certificate $C(x)$, and let a downstream consumer apply a validity test $V(x) \in \{0, 1\}$, where $V(x) = 1$ means the record supports the consumer's read. A certificate is *consumer-sound* when it predicts V :

$$C(x) = 1 \Rightarrow V(x) = 1. \quad (1)$$

A provenance certificate C_{prov} attests only that the source signed x and that x is unaltered. It is by construction independent of V , so

$$\Pr[V(x) = 0 \mid C_{\text{prov}}(x) = 1] = \Pr[V(x) = 0], \quad (2)$$

the base rate of invalid records. We call this conditional probability the *false-clear rate* of the certificate. A provenance certificate does not reduce it below the base rate.

When invalid records are common, which for raw observational data they are, the false-clear rate is high and the certificate is, in the theory's sense, vacuous. The failure is not weak cryptography. The cryptography is sound. The failure is that C_{prov} certifies the source, not the reader whose test is V .

V. THE VALIDITY LAYER

Four moves make a certificate consumer-sound, and each maps onto machinery that already exists.

Validity attestation, re-evaluated at use time. The certificate should attest that a consumer can validly extract, bound to calibration state and instrument envelope, and it should be re-evaluated when the data are used rather than fixed at capture. This is the validity-attestation class and use-time re-evaluation of IEEE P3787 [5].

A measured false-clear rate for the network. Of the records a system clears, the fraction that fail a real consumer test V is the actual false-clear rate of its certificate. Reporting this rate turns a trust claim into a measurement, and a high rate is a falsification.

Re-observation as the consumer test. For observational data the most consumer-relevant V is persistence: a signal that recurs under independent looks, and is absent where it should be absent, is supported in a way a single signed frame can never be. This is the ON-OFF cadence logic of technosignature search, promoted from a pipeline convenience to the definition of the certificate.

An in-band witness. The witness should live in the observation's own read path, so that it is coupled to the instrument's fidelity, rather than a separate anchor that certifies the record's history while decoupled from its measurement quality.

VI. EVIDENCE: A MEASURED CONSUMER TEST

We demonstrate the re-observation consumer on real data, to show that a validity test does work that a signature cannot. We ran the standard narrowband technosignature search on a real six-scan ON-OFF cadence of a nearby star, observed at L-band with the Green Bank Telescope and released through Breakthrough Listen [6], [7]. The raw search flags every narrowband hit in each scan. The cadence consumer keeps only a hit that appears in the on-source scans and is absent from the off-source pointings.

Across the three on-source scans the raw search returned 6651 hits, almost all of them attributable to Global Positioning System and satellite interference near 1575 MHz. The cadence consumer promoted 3 of them to candidates (Fig. 1). The false-clear rate of the raw single-scan detector, relative to the cadence consumer, is therefore $1 - 3/6651 \approx 0.9995$. A signature on any of these frames would have certified all 6651 as authentic and unaltered, and would have separated none of them. The discrimination comes from re-observation, not provenance.

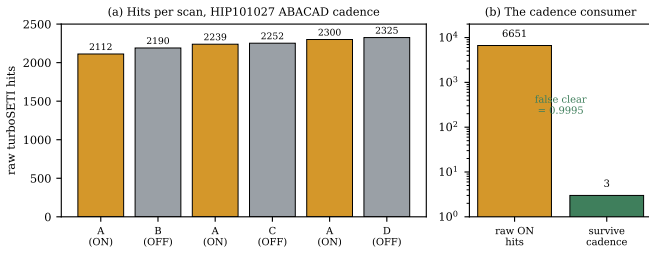


Fig. 1. The re-observation consumer on real data. (a) Raw narrowband hits in each scan of the six-scan ON-OFF cadence of HIP101027 (GBT L-band, MJD 57574, coarse channels 119–144). (b) Of 6651 raw on-source hits, the cadence keeps 3. One measurement, and 99.95% of the “detections” a signature would certify are gone.

We report the number as a mechanism demonstration, not as a benchmark of any deployed system. The tie to onchain anchoring is nonetheless exact at the level of the characterization. Take C_{prov} to be a provenance-only anchoring certificate and V the re-observation test the technosignature consumer applies. Under the pure-provenance reading of Section II, every one of the 6651 raw records anchors and clears identically, so $C_{\text{prov}} = 1$ on all 6651 while the consumer keeps 3. Equation (2) then fixes the certificate’s false-clear rate at the base rate $\Pr[V(x) = 0]$, which on this real stream is 0.9995. A provenance-only certificate over this observation stream false-clears at 0.9995, not because its cryptography is weak but because (2) pins it to a base rate that only re-observation moves. That rate is a property of what is anchored and which consumer reads it: anchor calibrated, cadence-screened products and it falls; anchor raw single-scan records, as the strongest-reading provenance-only design does, and it is this high.

VII. RELATED WORK

Blockchain proposals for scientific data provenance and reproducibility are numerous, and critical appraisals of them exist [1], [8]. Tamper-evident logging and trusted timestamping predate the ledger framing [3], [2]. The distinction between data provenance and data quality is long established in the databases and metrology literatures [9], [10], and our contribution is to carry it into the onchain-observation setting and to attach a measurable rate to it. Consumer-relative observation and the validity-attestation program provide the frame [4], [5]. We do not propose a better ledger. We propose the measurement that shows what a ledger does not do.

VIII. DISCUSSION AND SCOPE

The critique is narrow and precise. Provenance is necessary for trusted observational datasets. It is not sufficient. We do not claim that anchoring is worthless. For attribution and for verifiable pre-registration it is the right tool, and we say so in Section I.

We treat these systems at their strongest reading, as pure-provenance mechanisms. If a particular specification

incorporates instrument-level validity checks, the critique narrows from the design to the claim, and the constructive layer still applies. Our empirical demonstration uses one cadence, one frequency sub-band, and one narrowband consumer. It is a demonstration that a consumer test separates real data, not a survey-scale statistic, and we do not generalize it beyond that.

IX. CONCLUSION

Onchain anchoring proves that an observation is authentic. It does not prove that the observation is right, and for science whether it is right is the whole question. The validity layer, not the ledger, is where trust is earned. We have separated the two guarantees, characterized a provenance-only system as a false-clear mechanism with a base-rate false-clear rate, given the layer that closes the gap, and shown on real data that a consumer test delivers the separation a signature cannot.

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